

Q1. What is DAX? Why is it used?

DAX (Data Analysis Expressions) is a formula language used in **Power BI, SQL Server Analysis Services (SSAS) Tabular, and Power Pivot in Excel**. It is used to create custom calculations and analyze data in tabular models.

Why is DAX used?

- Create **calculated columns**
- Create **measures (aggregations)**
- Create **calculated tables**
- Perform time intelligence calculations (YTD, MTD, QTD)
- Apply filtering and business logic
- Build KPIs and dynamic reports

Example:

Total Sales = SUM(Sales[SalesAmount])

Q2. What are the data types supported by DAX?

DAX supports the following data types:

Data Type	Description
Whole Number	Integer values
Decimal Number	Floating-point numbers
Currency (Fixed Decimal Number)	Financial calculations with fixed precision
Date/Time	Dates and timestamps
Boolean	TRUE or FALSE
Text	Strings or characters
Blank	Represents null or missing values
Binary	Used internally (not commonly used in DAX formulas)

Example:

$\text{Profit} = \text{Sales}[\text{Revenue}] - \text{Sales}[\text{Cost}]$

Returns a numeric value.

Q3. What is the difference between a Calculated Column and a Measure?

Calculated Column	Measure
Computed row by row	Computed on demand
Stored in the model	Not stored; calculated at query time
Increases model size	Does not increase model size significantly
Uses Row Context	Uses Filter Context
Used for slicing, filtering, grouping	Used for aggregations and visual calculations

Calculated Column Example

$\text{Profit} = \text{Sales}[\text{SalesAmount}] - \text{Sales}[\text{Cost}]$

Each row gets its own profit value.

Measure Example

$\text{Total Profit} = \text{SUM}(\text{Sales}[\text{SalesAmount}]) - \text{SUM}(\text{Sales}[\text{Cost}])$

Returns a single aggregated value depending on filters in the report.

Key Difference

- **Calculated Column:** Calculated during data refresh.
- **Measure:** Calculated dynamically when the report is queried.

Q4. How to exclude all filters while calculating any measure?

Use the **ALL()** function (or **REMOVEFILTERS()** in newer DAX) inside **CALCULATE()**.

Example using ALL()

Total Sales =

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    ALL(Sales)  
)
```

This ignores all filters applied to the **Sales** table.

Remove filters from a specific column

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    ALL(Sales[Region])  
)
```

Using REMOVEFILTERS()

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    REMOVEFILTERS()  
)
```

Common filter functions

Function	Purpose
ALL()	Removes filters from specified tables or columns
REMOVEFILTERS()	Clears filters (recommended in newer DAX)
ALLEXCEPT()	Removes all filters except specified columns
ALLSELECTED()	Keeps filters selected by the user while removing others

Q5. What are circular dependencies? How can you avoid creating circular dependencies in your DAX expressions?

A **circular dependency** occurs when two or more DAX calculations depend on each other, creating an endless calculation loop.

Example

Column A = [Column B] + 10

Column B = [Column A] + 5

Here:

- Column A depends on Column B.
- Column B depends on Column A.

Power BI cannot determine which calculation should be performed first, so it throws a **Circular Dependency** error.

How to avoid circular dependencies

- Avoid creating calculated columns that reference each other.
- Use **measures** instead of calculated columns where appropriate.
- Break complex calculations into independent steps.
- Use variables (VAR) to simplify expressions and avoid repeated dependencies.
- Ensure relationships between tables do not create dependency loops.
- Avoid calculated columns that indirectly depend on themselves through relationships or other calculated columns.

Example using variables

Total Profit =

VAR SalesValue = SUM(Sales[SalesAmount])

VAR CostValue = SUM(Sales[Cost])

RETURN

SalesValue - CostValue

Using VAR makes the calculation clearer and reduces the likelihood of creating unnecessary dependencies.